

SATU MARE, Romania

WMO: 150100

Lat: 47.7214N

Lon: 22.8872E

Elev: 407

StdP: 14.48

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.3	12.6	2.2	6.2	8.7	8.4	8.6	15.2	22.8	34.9	20.2	34.1	3.2	90	0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	91.6	70.7	88.5	69.9	85.7	68.8	73.6	85.9	71.9	84.0	70.2	82.0	5.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.9	111.5	79.6	68.0	104.5	76.9	66.3	98.5	74.7	37.4	85.9	35.8	84.1	34.4	82.5	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.3	16.6	14.0	DB	0.5	95.1	9.2	3.2	-6.1	97.4	-11.5	99.3	-16.7	101.1	-23.4	103.4	(4)
(5)			WB	0.2	76.7	9.1	2.5	-6.4	78.5	-11.7	79.9	-16.8	81.3	-23.4	83.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	30.4	34.0	41.9	52.7	61.0	67.5	70.2	70.0	61.2	51.4	42.0	32.7		
	DBStd	16.00	8.81	9.17	7.79	6.91	6.25	5.84	5.46	5.82	6.80	7.53	8.22	8.54		
	HDD50	2253	609	448	264	50	4	0	0	0	2	76	262	538		
	HDD65	5528	1074	868	715	370	154	42	16	19	153	423	691	1003		
	CDD50	2739	0	0	14	132	344	525	626	620	337	120	21	0		
	CDD65	538	0	0	0	2	29	116	177	174	38	2	0	0		
	CDH74	6390	0	0	0	49	438	1254	2025	2071	521	32	0	0		
	CDH80	2358	0	0	0	8	88	408	813	881	159	1	0	0		
	WSAvg	5.4	5.1	5.7	7.0	6.5	6.1	5.4	5.0	4.6	5.0	4.7	5.2	5.0		
	PrecAvg	23.4	1.5	1.2	1.4	1.7	2.6	3.3	2.6	2.5	1.5	1.5	1.9	1.9		
(15) Precipitation	PrecMax	33.1	3.2	2.9	4.6	4.3	5.0	8.7	7.2	5.0	5.1	6.1	4.3	4.0		
	PrecMin	14.9	0.1	0.0	0.0	0.2	1.0	0.8	0.7	0.2	0.0	0.0	0.4	0.0		
	PrecStd	4.3	0.8	0.7	1.0	0.8	1.2	1.9	1.4	1.2	1.1	1.2	1.1	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	58.4	68.0	79.8	86.0	91.5	95.3	96.6	90.1	77.3	66.3	54.0		
		MCWB	47.3	49.4	53.5	61.4	66.1	71.9	71.6	70.1	67.2	61.5	55.4	49.1		
	2%	DB	48.1	53.2	62.9	73.7	82.5	87.9	91.3	92.4	84.2	73.6	62.2	50.1		
		MCWB	45.1	47.1	51.4	57.8	64.3	71.1	71.5	70.3	64.6	60.2	54.4	46.2		
	5%	DB	44.7	49.4	58.8	70.4	79.0	84.6	87.9	89.2	80.3	69.6	57.6	46.5		
		MCWB	42.6	45.1	49.1	56.1	63.3	68.9	69.8	69.9	63.8	59.1	51.9	44.3		
	10%	DB	41.3	46.1	55.0	66.8	75.4	81.3	84.5	85.0	75.4	64.6	54.0	43.2		
		MCWB	39.6	43.1	47.6	54.4	62.2	67.7	68.7	68.6	62.4	56.6	49.7	41.5		
	0.4%	WB	48.5	50.8	55.5	62.7	69.1	75.4	75.8	75.7	68.9	64.3	57.8	49.9		
		MCDB	50.6	55.9	64.2	76.4	80.2	86.5	87.2	87.6	83.0	72.4	63.3	52.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.3	47.7	52.4	59.9	66.7	72.9	73.3	73.4	67.0	61.7	55.2	46.5		
		MCDB	47.3	52.1	60.2	71.4	78.2	84.1	85.6	86.7	79.6	70.3	60.4	49.0		
	5%	WB	42.4	45.4	50.4	57.4	64.8	70.8	71.6	71.3	65.3	59.9	52.7	44.2		
		MCDB	44.1	48.6	57.2	67.5	75.7	81.3	84.8	83.2	76.5	67.6	57.4	46.3		
	10%	WB	39.8	43.0	48.2	55.5	63.1	68.8	70.0	69.7	63.4	57.8	50.4	41.6		
		MCDB	41.2	45.4	53.8	64.5	73.3	78.8	82.0	82.2	73.2	64.1	53.8	43.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	13.6	17.8	21.4	22.0	21.8	23.2	24.6	22.2	20.4	15.3	10.2		
		MCDBR	13.9	18.2	24.9	27.4	28.4	27.3	28.9	30.3	29.8	27.3	20.6	13.9		
		MCWBR	10.5	12.8	14.6	13.5	13.1	12.0	11.1	10.8	12.3	15.1	13.4	10.5		
		MCDBR	12.9	16.9	22.0	24.7	25.0	24.2	26.1	26.1	25.0	23.3	18.0	12.8		
	5% WB	MCWBR	10.2	12.3	13.7	12.7	12.5	11.5	11.0	11.5	11.4	13.6	11.9	10.2		
(40) Clear-Sky Solar Irradiance	taub		0.329	0.355	0.394	0.445	0.429	0.429	0.449	0.448	0.411	0.382	0.356	0.329		
	taud		2.289	2.251	2.204	2.097	2.209	2.239	2.196	2.194	2.262	2.314	2.330	2.328		
	Ebn at Noon		230	250	258	256	266	266	258	252	250	238	219	215		
	Edh at Noon		24	31	38	47	43	43	44	42	36	29	23	21		
(44) All-Sky Solar Radiation	RadAvg		321	581	968	1403	1717	1869	1833	1683	1170	745	406	238		
	RadStd		50	88	144	148	179	123	163	151	147	99	59	38		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.57	N/A	N/A	+2.36	N/A	N/A	-227	-371	+245	+122
(47)	Regional (48 neighbors)	+1.24	N/A	N/A	+2.00	+0.85	N/A	-247	-369	+205	+95

Nomenclature: See separate page